

Predictive factors for postoperative delirium in thoracic surgery.

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We examined cases in which delirium developed after thoracic surgery under general anesthesia at our hospital to determine the predictive factors for postoperative delirium, as well as the perioperative findings in cases showing postoperative delirium. This retrospective study included 1674 patients who underwent surgery under general anesthesia at our hospital between 2012 and 2022. A psychiatrist diagnosed postoperative delirium using the Confusion Assessment Method. There were 99 (5.9%) patients with postoperative delirium in our study, including 85 (86%) men, of whom 31 (31%) had a history of cerebrovascular disease. The incidence of postoperative delirium in patients aged > 80 years was 20% (36/182). The postoperative delirium group showed significantly longer hospital stays and more frequent postoperative complications than the group without postoperative delirium. In univariate analysis, age $\geq$ 80 years, male sex, history of cerebrovascular disease, hypertension, history of atrial fibrillation, and history of smoking were identified as significant factors, while multivariate analysis identified age $\geq$ 80 years, male sex, history of cerebrovascular disease, hypertension, and history of smoking as significant factors (odds ratios = 5.15, 2.04, 3.10, 1.67, and 2.36, respectively). In the 169 cases with none of these five factors, the postoperative delirium risk was 0% (0/169). In patients undergoing thoracic surgery, predictive factors for postoperative delirium include age $\geq$ 80 years, male sex, history of cerebrovascular disease, hypertension, and smoking history. The findings also indicate that patients with these risk factors may require psychiatric consultation before surgery.

Hotspot Exon 15 Mutations in BRAF Are Uncommon in Feline Tumours.

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BRAF is one of multiple RAF proteins responsible for the activation of the MAPK cell signalling cascade involved in cell growth, differentiation, and survival. A hotspot BRAFV600E mutation, in exon 15, was determined to be a driver in 100% hairy cell leukaemias, 50%-60% of human melanomas, 30%-50% of human thyroid carcinomas and 10%-20% of human colorectal carcinomas. The orthologous BRAFV595E mutation was seen in 67% and 80% of canine bladder transitional cell carcinomas and prostatic adenocarcinomas, respectively. Since veterinary and human cancers exploit similar pathways and BRAF is highly conserved across species, BRAF can be expected to be a driver in a feline cancer. Primers were developed to amplify exon 15 of feline BRAF. One hundred ninety-six feline tumours were analysed. Sanger sequencing of the 211 bp PCR amplicon was done. A BRAF mutation was found in one tumour, a cutaneous melanoma. The mutation was a BRAFV597E mutation, orthologous to the canine and human hotspot mutations. A common synonymous variant, BRAFT586T, was seen in 23% (47/196) of tumours. This variant was suspected to be a single nucleotide polymorphism. BRAF was not frequently mutated in common feline tumours or in tumour types that frequently harbour BRAF mutations in human and canine cancers. As is seen in canine cancer genomics, the mutational profile in feline tumours may not parallel the histologic equivalent in human oncology.

Identification of BRAF Inhibitor Resistance-associated lncRNAs Using Genome-scale CRISPR-Cas9 Transcriptional Activation Screening.

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Approximately 50% of melanomas harbor the BRAF V600E mutation and targeted therapies using BRAF inhibitors improve patient outcomes. Nonetheless, resistance to BRAF inhibitors develops rapidly and remains a challenge in melanoma treatment. In this study, we attempted to isolate long noncoding RNAs (lncRNAs) involved in BRAF inhibitor resistance using a comprehensive screening method. We used a CRISPR-Cas9 synergistic activation mediator (SAM) protein complex in a genome-scale transcriptional activation assay to screen for candidate lncRNA genes related to BRAF inhibitor resistance. Correlation analysis was performed between expression levels of isolated lncRNA genes and IC50 of dabrafenib in a BRAF-mutated melanoma cell line. Next, online databases were used to construct the lncRNA-miRNA-mRNA regulatory network. Finally, we evaluated the significance of the expression levels of these lncRNAs and mRNAs as biomarkers using clinical specimens. We isolated three BRAF inhibitor resistance-associated lncRNA genes, namely SNHG16, NDUFV2-AS1, and LINC01502. We constructed a lncRNA-miRNA-mRNA network of 13 nodes consisting of three lncRNAs, six miRNAs, and four mRNAs. The lncRNAs and target mRNAs from each regulatory axis significantly and positively correlated with each other. Finally, Kaplan-Meier analysis showed that higher expression levels of MITF, which was up-regulated by LINC01502, were significantly associated with worse prognosis in BRAF V600E-mutated melanoma. The identification of these BRAF inhibitor resistance-associated lncRNA genes at the genomic scale and the establishment of the lncRNA-miRNA-mRNA regulatory network provides new insights into the underlying mechanisms of BRAF inhibitor resistance in melanoma.

Therapeutic implications of the metabolic changes associated with BRAF inhibition in melanoma.

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Melanoma metabolism can be reprogrammed by activating BRAF mutations. These mutations are present in up to 50% of cutaneous melanomas, with the most common being V600E. BRAF mutations augment glycolysis to promote macromolecular synthesis and proliferation. Prior to the development of targeted anti-BRAF therapies, these mutations were associated with accelerated clinical disease in the metastatic setting. Combination BRAF and MEK inhibition is a first line treatment option for locally advanced or metastatic melanoma harboring targetable BRAF mutations. This therapy shows excellent response rates but these responses are not durable, with almost all patients developing resistance. When BRAF mutated melanoma cells are inhibited with targeted therapies the metabolism of those cells also changes. These cells rely less on glycolysis for energy production, and instead shift to a mitochondrial phenotype with upregulated TCA cycle activity and oxidative phosphorylation. An increased dependence on glutamine utilization is exhibited to support TCA cycle substrates in this metabolic rewiring of BRAF mutated melanoma. Herein we describe the relevant core metabolic pathways modulated by BRAF inhibition. These adaptive pathways represent vulnerabilities that could be targeted to overcome resistance to BRAF inhibitors. This review evaluates current and future therapeutic strategies that target metabolic reprogramming in melanoma cells, particularly in response to BRAF inhibition.

Population-based analysis of the prevalence of BRAF mutation in patients diagnosed with cutaneous melanoma and its significance as a prognostic factor.

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The prevalence of BRAF mutation has been reported in between 38% and 48% of melanoma patients, based on mainly Stage III or metastatic melanoma, however, information based on population-based studies is scarce. We performed a population-based retrospective cohort study to determine the prevalence of the BRAF mutation in patients diagnosed with in situ and infiltrating cutaneous malignant melanoma in the province of Girona between 2009 and 2011. Using the database of the Girona Cancer Registry, we performed BRAF mutation analysis based on paraffin-embedded tissue. This data was then correlated with other known clinical and histological prognostic factors for survival. We found 286 incident cases of cutaneous melanoma in the Girona Cancer Registry database. Excluding missing cases, BRAF-mutated patients constituted 38.9% of "in situ" melanoma cases and 53.8% of invasive melanoma cases. Five-year relative survival was not statistically different between BRAF-mutated patients (93.6%; 95% CI: 87.1-100.5) and non-mutated patients (84.3%, 95% CI: 75.3-94.8). Only stage was significant as a prognostic factor for survival based on multivariate analysis. From our population-based study, we conclude that BRAF mutation is not an independent prognostic factor for melanoma survival.

BRAF-V600E expression in primary nodular melanoma is associated with aggressive tumour features and reduced survival.

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Around 50% of primary melanomas harbour BRAF mutations, but their prognostic impact has not been clear. Recently, a BRAF-V600E mutation-specific antibody has become available for immunohistochemistry. Here, we investigated for the first time the prognostic impact of BRAF-V600E protein expression in primary melanoma. In a patient series of 248 nodular melanomas, BRAF-V600E and total BRAF expression were assessed by immunohistochemistry using tissue microarray sections of paraffin-embedded archival tissue. Mutation status was assessed by real-time PCR in cases with sufficient tumour tissue (n=191). Positive BRAF-V600E expression was present in 86 (35%) of the cases, and was significantly associated with increased tumour thickness, presence of tumour ulceration and reduced survival. Further, BRAF-V600E expression was an independent prognostic factor by multivariate analysis, whereas BRAF mutation status was not significant. There was 88% concordance between BRAF-V600E expression and mutation status. Our findings indicate that BRAF-V600E expression is a novel prognostic marker in primary melanoma.

Incidence of BRAF p.Val600Glu and p.Val600Lys mutations in a consecutive series of 183 metastatic melanoma patients from a high incidence region.

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Approximately 40-60% of melanomas from Caucasian populations carry activating mutations in the BRAF oncogene, with the most common being the p.Val600Glu (V600E) hotspot mutation in exon 15. The aim of the present study was to investigate the frequency of the less common p.Val600Lys (V600K) mutation in metastatic melanoma from a high incidence region. Dideoxy sequencing and fluorescent single strand conformation analysis were used to screen for mutations in exon 15 of BRAF in 183 cases of metastatic melanoma. The overall incidence of BRAF mutation (89/183, 49%) was very similar to other large studies of Caucasian populations. However, the frequency of the p.Val600Lys mutation was higher than in most other studies and comprised almost one-third of all BRAF mutations in our cohort (27/89, 30%). BRAF

p.Val600Lys mutations were present at a relatively high frequency in this cohort of metastatic melanoma patients (27/183, 15%). Assays used to screen for BRAF mutations in the clinic should be robust enough to detect the p.Val600Lys mutation, as this may have therapeutic implications.

Treating metastatic melanoma in 2014: what just happened and what is next?

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In recent years the expectation related to treating patients with metastatic melanoma has changed. Three years ago, a diagnosis of metastatic melanoma was bleak; only a small number of patients would benefit from standard treatment and the reason for treatment benefit was often unknown, leaving physicians with limited options to help their patients. Today, just 3 years later, there are proven therapies that allow physicians to expect to shrink tumors and extend the lives of their patients after diagnosis. In this article, Paul B. Chapman, MD, Memorial Sloan Kettering Cancer Center, summarizes the recent advances in the field of metastatic melanoma and looks ahead to topics such as focusing on specificity, shutting down the ERK pathway, intermittent dosing, discovering driver mutations, individualizing checkpoint inhibition, and identifying rejection antigens to identify what should occur next in order to continue to improve outcomes for a larger population of patients with metastatic melanoma. Dr. Chapman's paper reminds us of the exciting progress that has been made while focusing on the work that is left in the field. Dr. Chapman is a physician-scientist who specializes in diagnosing and treating melanoma that has metastasized to other parts of the body. Dr. Chapman led a clinical trial that paved the way, in part, to the approval of vemurafenib—a drug that targets a BRAF mutation that is present in approximately 50% of patients with metastatic melanoma. Dr. Chapman is one of five Memorial Sloan Kettering scientists to be appointed to a melanoma "Dream Team" that focuses on identifying potential therapies for metastatic melanoma patients who do not have the mutated form of the BRAF gene. Dr. Chapman is also a professor of medicine at the Weill Cornell Medical College and is the chair of the Melanoma Research Alliance Medical Advisory Board. In addition to the above, Dr. Chapman is a sought-after mentor for medical oncology fellows and has a current research interest in identifying novel means to more durably block signaling in the MAPK pathway in melanoma cells. Jedd Wolchok, MD, PhD, Scientific Program Committee Chair.

Frequency and spectrum of BRAF mutations in a retrospective, single-institution study of 1112 cases of melanoma.

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The US Food and Drug Administration (FDA) approved vemurafenib to treat patients with metastatic melanoma harboring the BRAF c.1799T>A (p.V600E) mutation. However, a subset of melanomas harbor non-p.V600E BRAF mutations, and these data are of potential importance regarding the efficacy of current targeted therapies. To better understand the BRAF mutation profile in melanomas, we retrospectively analyzed data from 1112 primary and metastatic melanomas at our institution. The cohort included nonacral cutaneous (n = 774), acral (n = 111), mucosal (n = 26), uveal (n = 23), leptomeningeal (n = 1), and metastatic melanomas of unknown primary site (n = 177). BRAF mutation hotspot regions in exons 11 and 15 were analyzed by pyrosequencing or with the primer extension MassARRAY system. A total of 499 (44.9%) specimens exhibited BRAF mutations, involving exon 15 [497

(99.6%)] or exon 11 [2 (0.4%)]. p.V600E was detected in 376 (75.4%) cases; the remaining 123 (24.6%) cases exhibited non-p.V600E mutations, of which p.V600K was most frequent [86 (17.2%)]. BRAF mutations were more frequent in non-acral cutaneous (51.4%) than acral melanomas [18 (16.2%)] ($P < 0.001$); however, there was no significant difference among cutaneous histological subtypes. All mucosal, uveal, and leptomeningeal melanomas were BRAF wild type (WT). The high frequency of non-p.V600E BRAF mutations in melanoma has important implications because the FDA-approved companion diagnostic test for p.V600E detects some but not all non-p.V600E mutations. However, the therapeutic efficacy of vemurafenib is not well established in these lesions.

Ursolic acid interaction with transcription factors BRAF, V600E, and V600K: a computational approach towards new potential melanoma treatments.

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Melanoma is one of the cancers with the highest mortality rate for its ability to metastasize. Several targets have undergone investigation for the development of drugs against this pathology. One of the main targets is the kinase BRAF (RAF, rapidly accelerated fibrosarcoma). The most common mutation in melanoma is BRAFV600E and has been reported in 50-90% of patients with melanoma. Due to the relevance of the BRAFV600E mutation, inhibitors to this kinase have been developed, vemurafenib-OMe and dabrafenib. Ursolic acid (UA) is a pentacyclic triterpene with a privileged structure, the pentacycle scaffold, which allows to have a broad variety of biological activity; the most studied is its anticancer capacity. In this work, we reported the interaction profile of vemurafenib-OMe, dabrafenib, and UA, to define whether UA has binding capacity to BRAFWT, BRAFV600E, and BRAFV600K. Homology modeling of BRAFWT, V600E, and V600K; molecular docking; and molecular dynamics simulations were carried out and interactions and residues relevant to the binding of the inhibitors were obtained. We found that UA, like the inhibitors, presents hydrogen bond interactions, and hydrophobic interactions of van der Waals, and π -stacking with I463, Q530, C532, and F583. The ΔG of ursolic acid in complex with BRAFV600K (-63.31 kcal/mol) is comparable to the ΔG of the selective inhibitor dabrafenib (-63.32 kcal/mol) in complex to BRAFV600K and presents a ΔG like vemurafenib-OMe with BRAFWT and V600E. With this information, ursolic acid could be considered as a lead compound for design cycles and to optimize the binding profile and the selectivity towards mutations for the development of new selective inhibitors for BRAFV600E and V600K to new potential melanoma treatments. The homology modeling calculations were executed on the public servers I-TASSER and ROBETTA, followed by molecular docking calculations using AutoGrid 4.2.6, AutoDockGPU 1.5.3, and AutoDockTools 1.5.6. Molecular dynamics and metadynamics simulations were performed in the Desmond module of the academic version of the Schrödinger-Maestro 2020-4 program, utilizing the OPLS-2005 force field. Ligand-protein interactions were evaluated using Schrödinger-Maestro program, LigPlot+, and PLIP (protein-ligand interaction profiler). Finally, all of the protein figures presented in this article were made in the PyMOL program.